PERFORMANCE EVALUATION

Objective

The objective of this phase was to evaluate the performance of the privacy-preserving iris authentication system. Different authentication, security, and computational metrics were analyzed to understand the effectiveness of the proposed system.

Authentication Metrics

1) **Accuracy**

   -> Accuracy measures the overall correctness of the authentication system.

   -> Result: Authentication Accuracy = 68.75%

   -> The system was able to correctly identify a large number of genuine and impostor authentication attempts.

2) **FAR (False Acceptance Rate)**

   -> FAR measures how often an unauthorized user is accepted by the system

   -> **Observation:** The FAR value was calculated using impostor matching scores.

   -> A lower FAR indicates better security.

3) **FRR (False Rejection Rate)**

   -> FRR measures how often a genuine user is rejected by the system.

   -> **Observation:** The FRR value was calculated using genuine matching scores.

   -> A lower FRR improves user convenience.

4) **EER (Equal Error Rate)**

   -> EER is an important biometric metric used to evaluate authentication performance.

   -> **Observation:** EER was calculated from FAR and FRR values.

   -> Lower EER indicates better system performance.

5) <u>ROC Curve</u>

-> A ROC curve was generated to visualize the authentication performance of the system.

-> **Observation:** Genuine scores were generally higher than impostor scores.

-> The selected threshold was able to separate most genuine and impostor attempts.

6) <u>Confusion Matrix</u>

-> The confusion matrix was generated using authentication results.

-> **Results:**

->> True Positive (TP) = 652

->> False Negative (FN) = 348

->> True Negative (TN) = 723

->> False Positive (FP) = 277

-> The confusion matrix helped in understanding the authentication decisions made by the system.

7) <u>Security Metrics</u>

-> The security of the proposed system was also evaluated.

-> **Results:**

	Metric	**Result**
0	Encryption Robustness	High
1	Reconstruction Resistance	Strong
2	Privacy Leakage	Low
3	Matching Reliability	Good

8) <u>Computational Metrics</u>

(i) <u>Encryption Time</u>

-> The time required for encrypting biometric templates was analyzed.

-> **Observation:** Encryption was successfully performed for all

templates.
-> Encryption time remained consistent throughout the process.

(ii) <u>Memory Usage</u>

-> The memory required for storing encrypted templates was evaluated.
-> **Observation:** Encrypted templates required more memory than normal feature vectors.
-> This increase is expected when using homomorphic encryption.

9) <u>Computational Overhead</u>

-> Homomorphic encryption increases the computational cost of authentication.
-> **Observation:** Matching was slower than traditional biometric matching.
-> The additional cost provides stronger privacy protection.

Comparative Analysis

	Method	Security	Accuracy	Speed
0	Raw Template Matching	Low	High	Fast
1	AES-Based Matching	Medium	High	Medium
2	Homomorphic Matching	Very High	68.75%	Slower

Conclusion

-> Successfully evaluated the performance of the proposed iris authentication system.
-> Achieved an authentication accuracy of 68.75%.
-> Calculated FAR, FRR, and EER metrics successfully.
-> Generated ROC curve and confusion matrix for performance analysis.
-> Verified strong security through encryption and privacy protection.
-> Analyzed encryption time and memory overhead.
-> Demonstrated that homomorphic encryption enables secure biometric authentication while preserving user privacy.